

When Model Selection Manufactures a Finding: A Tuning Trap in Quantum-Kernel Intrusion Detection

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Abstract. Selecting a hyper-parameter by cross-validated F_1 on class-imbalanced data can return a model that classifies every record into one class. We document a case in which this happened to a four-qubit quantum-kernel intrusion detector on UNSW-NB15, produced an apparent robustness result, and was refuted only when the regularisation constant was held fixed. We show the mechanism and show, by census, that it is not specific to the quantum kernel. We then measure what would have prevented it: on a denser grid the collapse does not survive selection under any of three objectives, but for one classical kernel every objective still returns a degenerate constant, so a class-averaged objective is a mitigation rather than a guarantee. The one check that identifies every case is the count of true negatives, and we release the checks that implement it.

Keywords: Quantum kernel · Model selection · Evaluation protocol · Intrusion detection · Reproducibility

1 Introduction

Quantum-kernel methods reached network intrusion detection quickly after quantum feature maps were proposed as a route to expressive kernels [2,3] and separations were established for constructed learning problems [4]; the applied literature now reports them on standard IDS corpora [10,11]. Asking what such methods actually deliver presumes that the measurement itself is sound. This paper is about a case where it was not, and where the failure was invisible in every number the protocol reported.

Evaluating a kernel classifier on intrusion-detection data means choosing a regularisation constant, and the standard way to choose it is to maximise a cross-validated score. On class-imbalanced data that procedure has a trap. The constant classifier that calls everything an attack attains a respectable binary F_1 , and that value acts as a floor beneath the whole grid. A model whose genuine scores never rise above the floor will have the degenerate setting returned to it by $\arg \max$, and will then be carried through the rest of the study as though it were a working model.

This happened to a four-qubit quantum-kernel detector in our own earlier work [1] on UNSW-NB15 [15]. The selected model predicted a single class for every test record. It went on to produce a robustness result — F_1 invariant under input perturbation — that was an artefact of the prediction being constant, and that result was not obviously wrong on inspection. It was caught only when the regularisation constant was fixed at a neutral value and the experiment re-run.

We report the case, the mechanism behind it, and a census showing that the collapse is a property of the interaction between the regularisation constant and the selection objective rather than of the quantum kernel. The contribution is not a better model; it is a caution about a procedure that most papers in this area run without reporting, and the checks that detect it.

Contributions.

1. A documented case in which $\arg \max$ selection over a cross-validated F_1 grid returned a degenerate classifier, and the degenerate classifier then produced an apparent robustness result.
2. The mechanism: on imbalanced folds the constant all-attack classifier attains $F_1 = 0.8504$, a floor that no grid point for the quantum kernel exceeded.
3. A census showing the collapse is a property of the regularisation-objective interaction rather than of the quantum kernel (Table 1).
4. A measurement, not a conjecture, of what would have prevented it: a denser grid removes the collapse for the quantum kernel under every objective tested, while a class-averaged objective leaves it in place for a classical kernel (Table 2). The two standard remedies are therefore partial, and the quantity that separates every case is the count of true negatives.
5. The audit checks that detect it, released with the paper.

2 Related work

Three lines of work meet here, and the gap this paper reports sits between them rather than inside any one of them.

Quantum kernels applied to intrusion detection. Quantum feature maps were proposed as a route to kernels that are hard to evaluate classically [2,3], and a separation was later established for a constructed learning problem [4]. The applied literature carried the idea to standard IDS corpora quickly: Payares and Martínez-Santos [10] compare quantum and classical models on denial-of-service detection, and Kalinin and Krundyshev [11] survey quantum techniques for intrusion detection. Both report aggregate scores — accuracy, precision, recall, F_1 — and neither reports a confusion matrix or any statement that the compared classifiers predict more than one class. Nothing in either paper suggests it suffered the failure reported here. The point is narrower and it concerns the reader: from what is printed, the failure would not be distinguishable from a working model.

Metrics that conceal a degenerate classifier. That binary F_1 flatters a classifier on imbalanced data is known. Chicco and Jurman [12] show that F_1 and accuracy return inflated scores on imbalanced problems precisely because neither reads all four cells of the confusion matrix, and argue for a coefficient that does. Christen *et al.* [13] gather the criticism of the F-measure and its alternatives in one place. In computer security the point is catalogued directly: Arp *et al.* [14] name *inappropriate performance measures* and the *base rate fallacy* among ten recurring pitfalls — the first being measures that ignore constraints such as imbalanced data, the second being imbalance ignored when results are interpreted.

What this paper adds is where the measure sits. That literature treats the choice as a *reporting* problem: the wrong number is printed about a model that exists independently of it. Here the same measure is the *selection* objective, so it does not misdescribe a model — it chooses one, and every number computed downstream describes a classifier that predicts a single class. A reporting error can be repaired afterwards by recomputing a metric on stored predictions. This one cannot, because the artefact is the model.

Evaluation blind spots that outlived a literature. The nearest precedent is not an intrusion-detection result at all. Guo *et al.* [7] showed that modern classifiers are systematically over-confident, and that the effect grows with capacity — a defect invisible in accuracy, which was the quantity the field reported, and obvious the moment a second quantity was plotted. The post-hoc corrections that followed [5] presume that diagnosis has been made. The present case has the same shape at smaller scale: one reported number, and one unreported number that decides whether the first means anything.

The constant C and the degenerate solution. The soft-margin formulation [6] trades margin width against hinge loss. As C falls the margin term dominates, and on a class-imbalanced sample the minimiser approaches the constant classifier that predicts the majority class — which on an attack-heavy split is the attack class. That much is standard and is not a defect of the solver. What this paper adds is that a selection objective can *prefer* that solution over every non-degenerate alternative on the grid, and that whether it does turns on a choice of averaging usually made without comment.

3 Setup

Both protocols use the same front end on UNSW-NB15: one-hot encoding, SELECTKBEST, PCA, and a min-max map onto $[0, \pi]$, every transformer fitted on the training subset alone. The quantum kernel is the ZZ-FeatureMap with full entanglement, evaluated as an exact statevector fidelity. Classical kernels are linear, degree-two polynomial, and RBF, all fitted by the same soft-margin solver [6] so that the only difference between arms is the kernel and the regularisation constant.

The evidence below comes from two protocols that must not be conflated. The *tuned* protocol uses 4 qubits, $N_{\text{train}} = N_{\text{test}} = 100$ and 5 seeds, with C chosen by

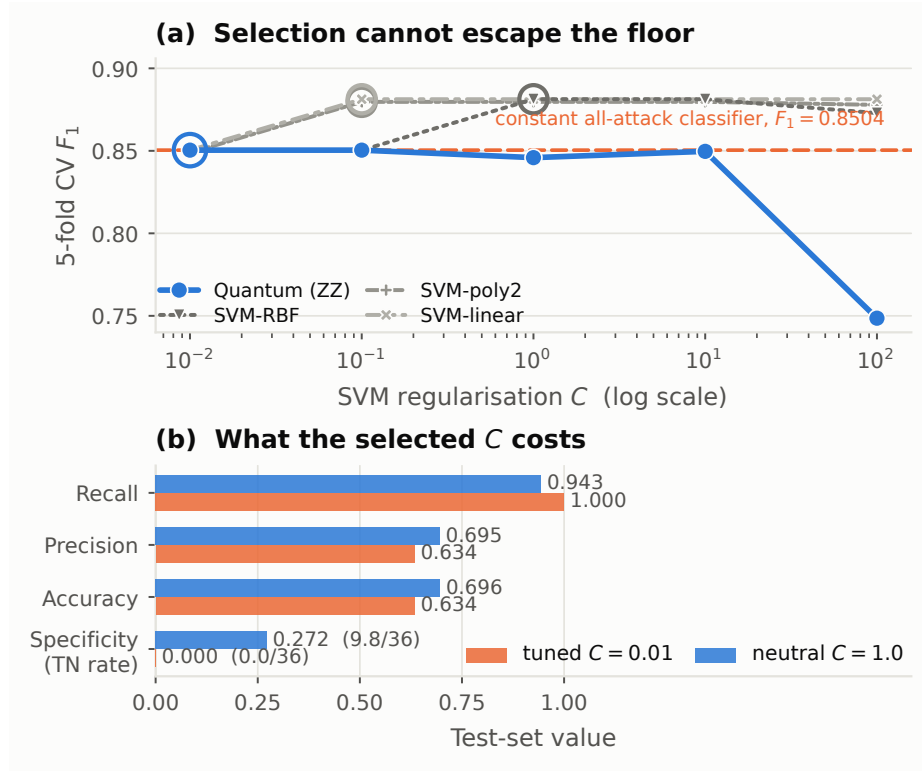


Fig. 1. (a) Cross-validated F_1 against C for four kernels. The dashed line is the F_1 of the constant all-attack classifier, 0.8504. All four kernels sit exactly on it at $C = 0.01$; the linear and polynomial kernels rise above it at $C = 0.1$ and the RBF kernel at $C = 1$, but the quantum kernel never does — its best non-degenerate score is 0.8497, below the floor. Circles mark the value $\arg \max$ returns. (b) What that selection costs on the test set.

$\arg \max$ over a 5-fold cross-validated F_1 on the grid $\{0.01, 0.1, 1, 10, 100\}$. The *revision* protocol uses 6 qubits, training sizes up to 10 000 and 10 runs. Every claim below states which protocol it belongs to.

4 The trap

On the tuned protocol the grid search returned $C = 0.01$ for the quantum kernel. The selected model has recall 1.000 with standard deviation 0 across all 5 runs, precision 0.634 equal to the attack ratio of the test split, and $TN = 0.0$: it calls every test record an attack.

The mechanism is Fig. 1(a). A constant all-attack classifier attains $F_1 = 0.8504201680672269$ on these folds. That value is a floor, and every kernel reaches it at $C = 0.01$. The linear and polynomial kernels climb above the floor once

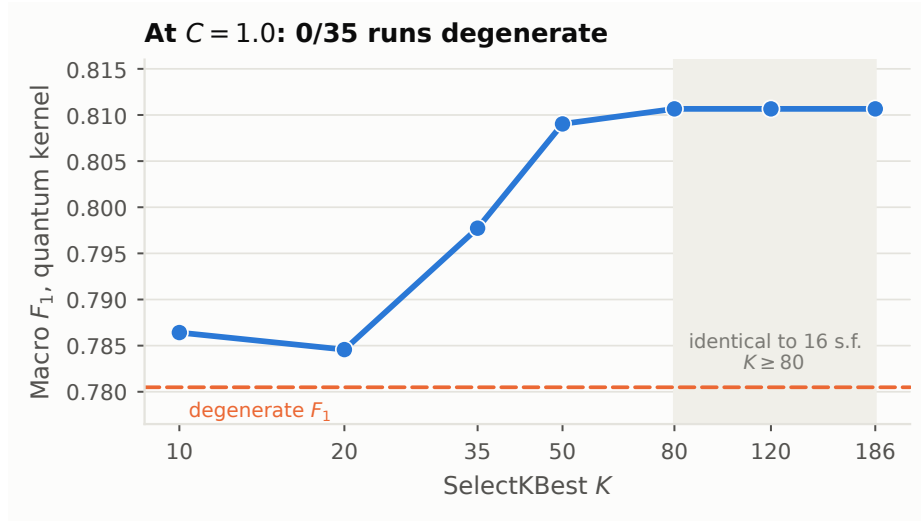


Fig. 2. With C held at a neutral 1.0, the quantum kernel is non-degenerate on 0 of 35 runs across seven feature budgets, and F_1 plateaus at 0.8107 for $K \geq 80$.

$C \geq 0.1$, so $\arg \max$ moves away from the degenerate point. The quantum kernel never climbs above it: its best non-degenerate grid score is 0.8497. Selection therefore had no non-degenerate maximum to find, and returned the collapse.

Two features of that account deserve to be stated as what they are: properties of the objective and of the grid, neither of which is a property of the kernel.

The objective ignores the evidence. Binary F_1 on the positive class does not read true negatives at all, so a classifier that never predicts the negative class loses nothing by it and is rewarded in proportion to the positive rate. Macro-averaged F_1 and balanced accuracy average over classes and score the constant classifier near 0.5. Whether that difference is enough to prevent the collapse is a measurable question rather than a rhetorical one, and Section 7 measures it. The answer is only partly the expected one.

The grid is complicit. The claim above that the quantum kernel never climbs above the floor holds for the grid that was searched, whose five points are spaced a decade apart. Section 7 shows that the interval in which this kernel does climb above the floor lies between two of those points, so the collapse survived selection partly because nothing was evaluated where the escape happens.

5 The kernel is not at fault

Refitting the same cached Gram matrices with $C = 1.0$ raises recall to 0.943, precision to 0.695 and TN to 9.8, and F_1 from 0.7760 to 0.7977. Kernel–target alignment is 0.19339506621329136 under *both* settings — identical to every stored digit. The Gram matrix did not change; only `SVC.fit` did.

Table 1. Degeneracy census. A model is degenerate on a run when it assigns every test record to the attack class ($TN = 0$). Collapse is not specific to the quantum kernel: under the tuned protocol it occurs for three of the four kernels, under the revision protocol only for an RBF kernel, and under a neutral C for none. What is specific to the quantum kernel is that it collapsed on *every* run, so no value of C on the grid scored above the degenerate floor and arg max selection locked the collapse in.

Protocol	Model	Selected C	Degenerate	Runs
tuned	Quantum (ZZ)	0.01	5	5
	SVM-linear	0.1	1	5
	SVM-poly2	0.1	1	5
	SVM-RBF	1	0	5
tuned data, neutral C	Quantum (ZZ)	1	0	35
revision	SVM-RBF	—	9	10

The apparent robustness result that the degenerate model produced is visible in the same artifacts: the spread of F_1 across perturbation strengths was 0.000 under the selected C — literally invariant, because the prediction was constant — against 0.0194 once C was held fixed.

6 A degeneracy census

Table 1 counts, for each protocol and model, the runs on which the fitted classifier assigned every test record to the attack class. Three points follow from it.

First, collapse is not a quantum phenomenon. Under the tuned protocol it occurs for three of the four kernels, the linear and polynomial kernels included. Under the revision protocol, whose model set also includes a random forest [9] and gradient-boosted trees [8], a scan of 840 rows across 7 models finds 9 degenerate rows, all belonging to SVM-RBF, and none belonging to the quantum kernel (whose lowest macro recall anywhere in that scan is 0.541).

Second, what is specific to the quantum kernel is not that it collapsed but that it collapsed on *every* run. That is what made the collapse survive selection: with no grid point scoring above the degenerate floor, arg max had no non-degenerate maximum available to it.

Third, holding the constant at a neutral value removes the effect entirely — 0 degenerate runs out of 35 across seven feature budgets — which is the evidence that the kernel was never the problem.

One point about provenance. The revision rows of Table 1 are recomputed from the runs the companion study [1] reports, but they count a quantity that study does not report: it states macro recall per run and never identifies a run as degenerate. The census is therefore a new measurement over existing runs, not a restatement of a published one, and we say so rather than leave it to be discovered.

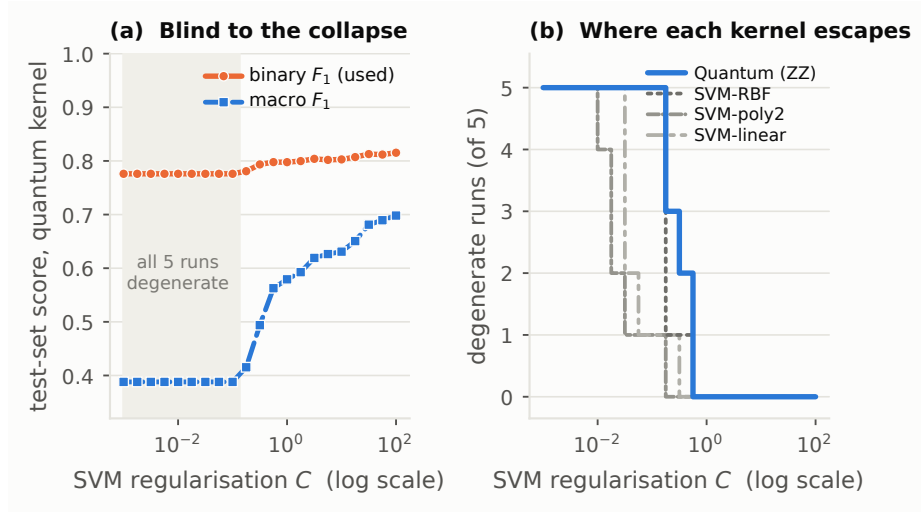


Fig. 3. (a) The quantum kernel across the collapse boundary, on the test set, averaged over 5 runs. Inside the shaded band every run assigns all test records to the attack class. Binary F_1 — the quantity selection maximised — moves by $+0.0218$ across the band edge; macro F_1 moves by $+0.1747$. (b) Degenerate runs against C for all four kernels. The quantum kernel escapes between 0.1 and 1, where the original grid had no point.

7 What would have caught it

Section 4 named two things that separate the run that collapsed from one that would not have: the averaging in the selection objective, and the grid over which it was maximised. Both were assertions. This section measures them.

The measurement reuses the stored Gram matrices of the tuned protocol, so no quantum circuit is evaluated again and the models compared are the same models. We refit on the cached kernels at 21 values of C spaced logarithmically from 0.001 to 100 — the original five points are a subset — on all 5 runs, and we repeat the cross-validated selection under three objectives. Before writing any new number, the same code path reproduces three published ones: the cross-validated F_1 at the original five grid points agrees to 0, and the test-set means at the tuned and at the neutral C agree to $2.2\text{e-}16$. A pipeline that could not return the numbers already in print would not be entitled to produce new ones.

One objective is nearly blind. Fig. 3(a) follows the quantum kernel from the degenerate floor to the smallest C at which no run collapses, $C = 0.562341$. Across that transition the classifier stops predicting a single class — the largest qualitative change it can undergo. Binary F_1 records it as $+0.0218$ (0.7760 to 0.7978). Macro F_1 records it as $+0.1747$ (0.3880 to 0.5627), about 8 times larger for the same event. That ratio is the quantitative content of the remark that

Table 2. What selection returns on a denser grid, under three objectives. The grid is the same for every row; only the quantity $\arg \max$ maximises changes. *Degenerate* counts the runs on which the selected model assigns every test record to the attack class. Two things follow. A class-averaged objective is not sufficient: for the linear kernel all three objectives return the same C and that C is still degenerate on one run. And the original grid, not only the objective, was part of the failure: on this grid even binary F_1 finds a non-degenerate maximum for the quantum kernel.

Kernel	Selection objective	Returns C	CV score	Degen.	Macro F_1
Quantum (ZZ)	binary F_1	0.562	0.8692	0/5	0.5627
	macro F_1	10	0.6744	0/5	0.6308
	balanced accuracy	10	0.6762	0/5	0.6308
SVM-RBF	binary F_1	0.316	0.8813	1/5	0.5422
	macro F_1	31.6	0.6317	0/5	0.6291
	balanced accuracy	31.6	0.6333	0/5	0.6291
SVM-poly2	binary F_1	17.8	0.8870	0/5	0.6285
	macro F_1	17.8	0.6423	0/5	0.6285
	balanced accuracy	17.8	0.6400	0/5	0.6285
SVM-linear	binary F_1	0.0562	0.8813	1/5	0.5669
	macro F_1	0.0562	0.6216	1/5	0.5669
	balanced accuracy	0.0562	0.6200	1/5	0.5669

binary F_1 ignores true negatives. It does not miss the collapse entirely; it under-reports it by close to an order of magnitude, which is the more dangerous failure, because a number that moves a little looks like a number that is working.

The grid mattered as much as the objective. Table 2 gives what $\arg \max$ returns on the denser grid. For the quantum kernel *every* objective now returns a non-degenerate constant, including the binary F_1 that failed originally: it returns $C = 0.562$ with 0/5 runs degenerate. Fig. 3(b) shows why. This kernel escapes between 0.1 and 1, and the original grid stepped straight across that interval. The collapse was therefore not the objective’s doing alone; adding points where the escape happens would have avoided it with the objective untouched.

A class-averaged objective is not a guarantee. The obvious remedy fails on one of the four kernels here. For the linear kernel all three objectives return the same constant, $C = 0.0562$, and it is degenerate on 1/5 runs. Class-averaged objectives do lower the score of a constant classifier, but that is not the same as making a degenerate solution unreachable: they change which model scores highest, not whether the highest-scoring model predicts two classes. On the RBF kernel the remedy does work — binary F_1 returns a constant degenerate on 1/5 runs, while both class-averaged objectives return $C = 31.6$ with 0/5. The remedy is real and partial. A study that adopted it and reported nothing further could still publish a constant classifier.

What separates the rows that collapsed from the rows that did not is not the objective. It is a single number that no aggregate score in Table 2 displays: the count of true negatives.

8 The checks

The fourth contribution is a set of checks, and they are worth describing because they are small. Three of them run over this paper itself.

The first recomputes every number in the text from the stored artifacts. Numbers reach the prose only through generated macros, and a separate scan fails the build if a literal decimal appears in a sentence. This is what makes the paper auditable rather than merely careful: a reader who doubts a number can find the file it was computed from.

The second asserts the invariants the argument rests on, so that the argument fails loudly if an artifact is regenerated and the conclusion no longer follows from it. Two of them: on the original grid the quantum kernel’s best non-degenerate cross-validated score must lie *below* the degenerate floor, and each classical kernel’s best score must lie above it. If either reversed, Section 4 would be wrong, and the check says so before a reader has to.

The third scans for degeneracy without filtering first. The census of Table 1 is computed over all 7 models and 840 rows of the revision protocol, not over the models we expected to find it in. That is what makes its negative result — no quantum-kernel row is degenerate there — worth reporting at all.

None of this is specific to quantum kernels or to intrusion detection, and the operational content is two lines in a reporting routine: record the confusion matrix beside the score, and assert that each fitted model predicts more than one class before any downstream result is computed. The case in this paper would have been caught at the moment it was created.

9 Limitations

The degeneracy evidence is a 4-qubit, $N = 100$, 5-seed result. It has *not* been reproduced under the 6-qubit, ten-run revision protocol, whose symmetric tuning procedure was adopted partly in response to this finding. The census in Table 1 therefore compares two protocols that differ in more than one respect, and we draw no conclusion from that comparison beyond the presence or absence of collapse.

The case is drawn from one corpus and one model family, and the degenerate protocol used a small training and test split, so the census counts runs rather than estimating a rate.

Section 7 carries its own limit. It measures what a denser grid and two other objectives would have returned *here* — same corpus, same 5 runs, same $N = 100$ split, same cached kernels. It therefore says what those choices would have done in this case, and not what they do in general; the linear-kernel row is enough

to show that the general statement would have been false. We do not claim the trap is common in the literature, only that it is easy to fall into and invisible in the reported numbers when it happens.

10 Conclusion

Selecting a regularisation constant by maximising a cross-validated binary F_1 can return a classifier that predicts one class for every input, and the resulting model will pass through an evaluation pipeline producing numbers that look like findings. We documented one such case, identified the mechanism — a degenerate floor in the selection objective that the model’s genuine scores never cleared — and showed by census that the collapse belongs to the regularisation-objective interaction rather than to the quantum kernel.

We then measured what would have prevented it, rather than asserting it. On a denser grid the collapse does not survive selection for the quantum kernel under any of three objectives, including the binary F_1 that failed originally — so the grid was as much at fault as the objective. But for the linear kernel all three objectives still return a degenerate constant, which makes a class-averaged objective a mitigation and not a guarantee.

One check identified every degenerate case in this paper, and it is the cheapest of them: report the confusion matrix beside the score. A constant predictor is then impossible to miss — the same reflex that surfaced systematic over-confidence in deep classifiers [7], and that a recalibration step [5] presumes has already been taken. We release the checks that implement it.

Reproducibility. TODO: dia chi that + release tag.

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